Hypergraph BiFormer for Semantic Segmentation of High-Resolution Remote Sensing Images

Weipeng Jing, *Member, IEEE*, Wenjun Zhang, Donglin Di, Chao Li, Mahmoud Emam, and Ajmal Mian, *Senior Member, IEEE*

***Abstract*— While transformers are powerful neural network architectures for feature learning, current Transformer-based approaches for semantic segmentation of high-resolution remote sensing images (HRRSIs) struggle with the extraction of local semantic features. To address this issue, we incorporate a hypergraph into the Transformer. Hypergraph-based methods are proficient at discovering high-order correlations within limited-scale data, extracting pertinent representations to enhance the Transformer's learning capabilities. We also propose dual pooling and feature aggregation modules (FAMs), inspired by the adaptive pooling's potent local modeling capabilities, to additionally extract fine-grained features from HRRSIs. In particular, we conceive a hypergraph BiFormer (HGBT) based on these three proposed modules along with a BiFormer backbone. HGBT has the potential to learn general latent features as well as generate high-order representations of HRRSIs by modeling correlations of multiscale features and local topology within an entirely nonlinear space, leading to the aggregation of features in a compact and localized manner, enhancing the model's ability to capture detailed variations within small areas. We validate our approach through extensive experiments on ISPRS Vaihingen and Potsdam datasets, where HGBT attains mean intersection over union (mIoU) of 83.71% and 87.88%, respectively. Both quantitative and qualitative assessments underscore the dominance of HGBT. Our code will be accessible at: https://github.com/ZhangIceNight/HGBFormer.**

***Index Terms*— Deep learning, hypergraph, remote sensing (RS), semantic segmentation.**

I. Introduction

With the continuous development of spatial observation and remote sensing (RS) technologies, high-resolution RS imagery is increasingly garnering widespread attention and finding diverse applications across various fields, such as land cover classification [1], urban planning and monitoring [2], as well as tree species classification [3]. These applications heavily rely on the abundant geospatial information furnished by the RS technology. However, accurately extracting regions of interest from the vast volumes of RS data remains a challenging problem. Therefore, research focused on semantic segmentation of high-resolution RS imagery holds significant importance.

High-resolution RS images (HRRSIs) of complex scenes contain a diverse range of objects with varying sizes, shapes, and spectral properties. These objects belong to different categories, such as buildings, grasslands, and shrubs. Notably, the objects in these images exhibit multiscale characteristics, and their spatial arrangement is irregular, leading to intricate and often complex spatial relationships. However, as RS image resolution continues to improve, and the complexity of RS image scenes continues to increase, enhancing the capability of segmentation methods to capture these complex relationships becomes a formidable challenge. It is precisely due to these complexities that the performance of conventional methods falls short of meeting the demands of contemporary applications [4]. This limitation significantly restricts the widespread and effective application of RS imagery across various fields. Consequently, the task of HRRSI semantic segmentation urgently requires innovative approaches to address these issues.

Convolutional neural networks (CNNs) have demonstrated their effectiveness in obtaining local contextual information [5], [6], [7]. The fully convolutional network (FCN) discards the fully connected layers, achieving a significant breakthrough in semantic segmentation accuracy through the integration of convolutional operations and skip connections. This advancement, however, gives rise to challenges that encompass two distinct facets: 1) hierarchical features generated through pooling layers can lead to a partial loss of localization and 2) due to their inherent fully convolutional nature, FCNs lack the inherent capability to encapsulate global contextual information within an image. Consequently, to alleviate the constraints that impede the efficacy of FCNs in RS domains, methods within the U-Net series [8], [9], [10] devised fully symmetrical encoder–decoder architectures, incorporating skip connections within their framework.

After the great success of Transformer-based methods in natural language processing (NLP) tasks [11], [12], transformers have also made a mark in the field of computer vision [13], [14]. Transformer-based methods [15], [16], [17], [18], [19], [20] have significantly alleviated the predicament of extracting global contextual information in semantic segmentation tasks. As illustrated in Fig. 1(b), unlike CNNs,

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Transformer-based methods divide the images into several patches and treat them as sequential data. However, when dealing with high-resolution RS imagery, these methods face challenges in capturing local features and performing multiscale modeling, which are crucial for successful segmentation. Transformer-based methods are not effective in obtaining local features of the image due to its sequential structure [21], [22]. Extracting local features for small target recognition from subdivided patches becomes challenging due to the high information density in RS images. Therefore, these methods may need further consideration on how to more effectively capture local details in the image, as well as how to better model complex semantic relationships when dealing with HRRSIs.

To address these pressing challenges, we propose a novel hierarchical network framework, namely, hypergraph BiFormer (HGBT) for semantic segmentation of HRRSIs. Specifically, we incorporate hypergraph convolution and dual-adaptive pooling modules into BiFormer to alleviate the deficiencies of BiFormer in local context modeling. With our modules, HGBT can model correlations that are more complex than pairwise relationships. Moreover, we develop a novel transformer block to construct a BiFormer backbone and an adaptive pooling block for local dependencies modeling.

As shown in Fig. 1(c), a dual-pooling module (DPM) with adaptive pooling [23] and soft pooling [24] is employed to capture local information in both the vertical and horizontal dimensions. As shown in Fig. 1(d), the hypergraph is a generalization of simple graphs and can model more complex geometric topology between multiple vertices in addition to capturing local feature correlations like a simple graph. Previous works [25], [26], [27] have already proven the effectiveness of hypergraph structure. Furthermore, as shown in Fig. 1(e) and (f), the FAM significantly bolster multiscale feature extraction capabilities, effectively equipping the model to handle a diverse range of challenges, including the segmentation of small targets and the complex background scenarios often encountered in HRRSIs.

In summary, our main contributions are as follows.

1) We propose a novel HGBT, which incorporates a hypergraph convolution module (HGM) designed to enhance the local contextual information capturing capability of the Transformer. The proposed HGM enhances the Transformer by establishing the topological structure and capturing high-order information.
2) To address the challenge of detecting small targets in HRRSIs, a DPM is designed to expand the receptive field and complement the information discarded by BiFormer blocks. Moreover, we propose a feature aggregation module (FAM) to extract multiscale features and detect small targets in the background.
3) We conduct extensive experiments on two ISPRS datasets. The proposed HGBT consistently outperforms state-of-the-art methods across the board and improves the mean intersection over union (mIoU) to 83.71% on the ISPRS Vaihingen dataset, an increase of 2.37%, and 87.88% on the Potsdam dataset, an increase of 1.44%.

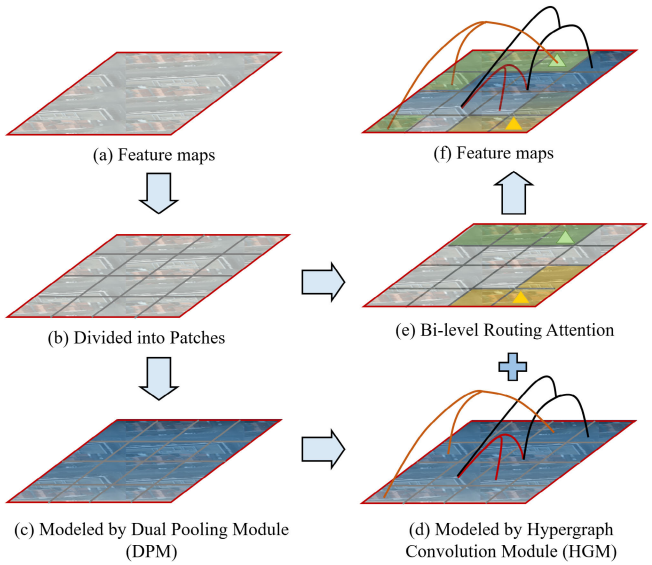


Fig. 1. Illustration of the proposed HGBT block. (a) RS image or the input features. (b) Features are divided into patches. (c) Proposed DPM. (d) Proposed HGM. (e) BRA with the proposed FAM. (f) Output features.

II. Related Works

A. CNN-Based Methods

Both conventional clustering and thresholding approaches [4] as well as methods that rely on fully connected layers [28] have demonstrated unsatisfactory accuracy in the segmentation performance. As a response, FCNs [29] were proposed to enhance prediction accuracy and ameliorate inference speed by adopting a structural paradigm devoid of fully connected layers. This involves the introduction of a skip connection-based architecture, enabling the establishment of connections between nonadjacent layers. Such an approach significantly improved segmentation performance. The incorporation of FCNs marks a pivotal landmark in semantic segmentation research. A major proportion of contemporary state-of-the-art methods essentially stem from the foundational framework established by FCN.

An assemblage of methods grounded in FCNs [30], [31], [32] have been introduced with the aim of addressing problems inherent to semantic segmentation. Pastorino et al. [33] leveraged the synergy of FCNs and feedforward neural networks to harness multiscale information, enhancing classification accuracy, particularly in scenarios with limited labeled training data. It was the integration of spatial–contextual information through additional loss terms that further improved the results. Pastorino et al. [34] proposed an approach for semantic segmentation of synthetic aperture radar images by combining FCNs, hierarchical probabilistic graphical models, and decision tree ensembles to integrate long-range spatial information and multiresolution features. The leveraging superpixel segmentation to effectively guide FCNs, Chen et al. [32] achieved rapid pixel-level segmentation without manual labels.

FCNs were soon outperformed by the encoder–decoder structure, notably recognized as the U-Net [35] and SegNet [36], as well as the feature pyramid network (FPN). Numerous methodologies built on the U-Net architecture have been employed for RS images [37], [38], [39], [40], show-

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casing remarkable adaptability and enhanced segmentation precision. Zheng et al. [41] introduced an innovative approach for accurate building extraction from aerial images by proposing an end-to-end edge-aware network (EANet) that focuses on low-level edge details often overlooked by previous methods. Liu et al. [38] combined U-Net with edge-guided and minority categories extraction branches to emphasize edge information and leverages multiscale contextual cues.

Despite the considerable advantages of CNNs in local feature extraction and multiscale representation, they still suffer from insufficient global information due to the inherent limitation of their receptive fields. In recent years, the emergence of attention mechanisms and Transformers has substantially augmented the capability of deep learning models to effectively capture and represent global features.

B. Attention-Based Methods

The attention mechanism has found increasing use in RS semantic segmentation tasks. This can be succinctly categorized into what to allocate attention to (channel attention), where to direct attention (spatial attention), when to focus attention (temporal attention), and the amalgamation of these aspects [42]. Su et al. [43] introduced a wavelet frequency channel attention block, enhancing the feature extraction by considering both spatial- and frequency-domain details. Zhi-min et al. [44] improved semantic segmentation of RS images by enhancing class correlations, spatial context extraction, and refining results using an encoder–decoder structure. Ramachandran et al. [45] introduced the use of local self-attention to replace spatial convolutions in a CNN to further explore the integration of local self-attention with convolutional operations. A horizontally connected residual blocks-based multiscale attention network is proposed by Li et al. [46] to enhance building extraction quality in RS images through feature discrimination, contextual understanding, and multilevel information integration. Fan et al. [47] incorporated a central composite attention mechanism to capture global river information dependencies, enhancing segmentation accuracy by dynamically combining binary cross-entropy and Dice coefficient losses. Fu et al. [48] adaptively integrated local features through two attention modules, enhancing both spatial and channel interdependencies.

C. ViT-Based Methods

The Transformer is an efficient architecture composed of a sequence of consecutive multihead self-attention layers, which demonstrated remarkable success in the field of NLP [11], [12]. With the extensions undertaken by numerous researchers [15], [16], [49], [50], [51], it has been substantiated that the Transformer can similarly achieve commendable performance within the domain of computer vision, which is evident across diverse tasks including classification [17], object detection [52], and semantic segmentation [53], [54].

The vision Transformer (ViT) [13] introduces a groundbreaking approach by splitting images into patches and incorporating linear position embeddings, constituting a pure Transformer architecture in this context. ViT processes these patches in a manner akin to how words are handled in NLP tasks. Following pretraining on expansive datasets, ViT showcases a simplicity that scales effectively, thereby paving the way for a multitude of vision applications. The segmentation transformer (SETR) [55] introduces a novel perspective by treating semantic segmentation as a sequence-to-sequence task, utilizing a pure transformer for encoding images into patch sequences. Pyramid ViT (PVT) [56] achieves high-resolution outputs while minimizing computational costs through dense partition training and a shrinking pyramid approach. Lu et al. [57] proposed a weakly supervised method with pseudolabels achieving segmentation results close to fully supervised methods. PVT combines the strengths of both CNN and Transformer, offering a convolution-free unified backbone for various vision tasks and serving as a direct substitute for traditional CNN backbones. SegFormer [18] combines Transformers with lightweight multilayer perceptron (MLP) decoders and introduces a novel hierarchical Transformer encoder that generates multiscale features without the need for positional encoding, ensuring performance consistency across different resolutions. By employing the simple yet effective MLP decoder, SegFormer combines local and global attention for powerful representations. Point cloud transformer (PCT) [58] adapts the Transformer architecture for point cloud segmentation. While ensuring the rotational and permutation invariance of point clouds, PCT incorporates enhanced input embedding methods, such as farthest point sampling and nearest neighbor search, which contribute to an enhanced capture of the local context within the point cloud structure.

Despite the extension of ViT to downstream tasks involving dense predictions like semantic segmentation, the challenges of high computational complexity and low feature resolution persistently present hurdles that are yet to be fully addressed. Consequently, subsequent endeavors have been focused on offering diverse solutions to mitigate these intricate issues [19], [20], [59], [60], [61], [62], [63]. For instance, the Swin Transformer [19] achieves the computation of self-attention within nonoverlapping local windows, thereby attaining linear computational complexity. However, while this approach represents a significant advancement, it still faces inherent challenges in effectively capturing intricate local features and performing adept multiscale modeling, particularly when confronted with high-resolution RS imagery. In response to these challenges, recent advancements have given rise to innovative variants. The cross-shaped window (CSWin) Transformer [59] and the DCSwin Transformer [60], for instance, endeavor to elevate the acquisition of multiscale features by introducing novel design choices for windows or integrating dense connections. These novel variants seek to fortify the model's capability to effectively encompass and harness the diverse intricacies inherent in RS imagery at varying scales. BiFormer [20] employs the bilevel routing attention (BRA) design to selectively filter out patches with weaker correlations to the current patch. This strategic filtering significantly reduces the computational workload of self-attention calculations. ST-UNet [61] incorporates the auxiliary encoder, which consists of Swin Transformer blocks, facilitates the provision

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of global context information to the main encoder based on the CNN architecture.

D. Hypergraph-Based Methods

Hypergraphs, as a generalization of traditional graphs, allow edges to connect multiple vertices simultaneously, enabling more complex and higher order relationships to be modeled. Unlike traditional graphs that can only capture pairwise relationships, hypergraphs can represent the inherent high-order correlations among multiple vertices, making them particularly suitable for complex visual tasks [27], [64]. Recent studies have demonstrated the effectiveness of hypergraph-based methods in various computer vision applications, including image segmentation [65], feature learning [26], and semantic understanding [66]. In the field of RS, hypergraph-based approaches have achieved remarkable success in various challenging tasks. These include image segmentation, where hypergraphs effectively model the complex spatial–spectral relationships [67], multimodal image retrieval through light-weighted hypergraph neural networks [68], change detection by leveraging multiview hypergraph fusion [69], hyperspectral image classification using feature fusion hypergraph [70], and road extraction using dual convolutional networks with hypergraph modeling [71]. These applications demonstrate the powerful capability of hypergraphs in capturing intricate dependencies and high-order relationships in RS data analysis.

Building upon the insights gained from the aforementioned studies, we present the integration of the HGM and DPM. These are coupled with BiFormer blocks enhanced by a proposed FAM, thereby enabling the simultaneous extraction of both high-quality global and local contextual information.

III. Methods

This section outlines the proposed approach, beginning with a brief summary of the overall architecture of the HGBT and an explanation of the BiFormer component. We then propose the HGM, which is named HGM. Finally, we further show the two important pooling modules in HGBT, which are named DPM and FAM.

A. Overall Architecture

As shown in Fig. 2, our approach adheres to the contemporary advancements in ViTs, employing a stratified four-stage pyramid architecture. In the first stage, we employ an overlapped patch embedding strategy to ensure that adjacent objects of the same category appear in the same cropped RS image as much as possible, followed by the integration of a patch merging module in the remaining stages. This amalgamation curtails the spatial resolution of the input image while concurrently augmenting the channel dimensions. The proposed HGM and DPM help in resolving the semantic ambiguity due to occlusion. In addition, we propose the FAM to integrate an innovative patch merging methodology, which enables the model to capture small-scale objects.

Given an RS image $\mathbf{X} \in \mathbb{R}^{H\times W\times 3}$, the patch embedding layer encodes the image into an appropriate size and feature dimension to strike a balance between accuracy and computational cost in stage 1, which is $[(H/4), (W/4), 2C]$. In stages 2–4, the patch merging layer with FAM reshapes and merges the input to $(1/4)$ of the input size, and the dimension is doubled. HGBT Blocks are repeated $[4, 4, 18, 4]$ times in stage $[1, 2, 3, 4]$, respectively. Finally, a feature map with shape and dimension in $[(H/32), (W/32), 8C]$ is sent to a segment decoder to predict a dense mask. To ensure a fair comparison, we keep consistent with BiFormer and other common methods utilizing UperNet [72] frameworks.

B. BiFormer

Supposing $\mathbf{X}$ and $\mathbf{X}' \in \mathbb{R}^{N\times C}$ as the input and output of the current Transformer block and $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ as the attention vectors generated by $\mathbf{X}$, respectively, a standard attention function and a Transformer block can be formally defined in a compact matrix form, as follows:

$$\begin{aligned}\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) &= \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{C}}\right)\mathbf{V}\\ \mathbf{X}' &= \text{MLP}(\text{LN}(\text{Attention}(\cdot))) + \text{Attention}(\cdot).\end{aligned} \tag{1}$$

A conventional Transformer has a complexity of $O(N^2)$ because each of the N queries must attend to N key–value pairs. This presents major scalability challenges, especially for dense prediction tasks with high-resolution inputs. For efficient modeling, the BiFormer proposed a dynamic, query-aware sparse attention mechanism, namely, BRA mechanism. They performed self-attention only in a small portion of routed regions by region-to-region routing with a directed graph, where each patch computes attention weights only with its k neighbors. To keep the comparison fair and consistent, we adhere to the same setting of k value as in BiFormer-S. As illustrated in Fig. 1, BRA addresses the scalability problem of standard Transformers by filtering out the most irrelevant key–value pairs at a coarse regional level.

Different from standard attention, BRA computes the semantic relationship between two regions and identifies the top-k indices $\mathbf{I}_k$ of the most relevant regions after deriving the region vectors $\mathbf{Q}^{\mathbf{r}}$, $\mathbf{K}^{\mathbf{r}}$, and $\mathbf{V}$ by

$$\mathbf{I}_k = \text{topk}\left(\mathbf{Q}^r(\mathbf{K}^r)^T\right) \tag{2}$$

where $\mathbf{Q}^r$, $\mathbf{K}^r$, $\mathbf{V} \in \mathbb{R}^{S^2\times C}$, and $\mathbf{I}_k \in \mathbb{N}^{S^2\times k}$, and S^2 and C represent the number and dimension of patches, respectively. Therefore, we can briefly express the attention map of BRA using the following representation:

$$\mathbf{O} = \text{Attention}\left(\mathbf{Q},\mathbf{K}^g,\mathbf{V}^g\right) \tag{3}$$

where $\mathbf{K}^g$ and $\mathbf{V}^g$ are the gathered key and value vectors.

C. Hypergraph Convolution Module

As presented in Fig. 3, a hypergraph is the generalization of a simple graph. In a hypergraph, two vertices can be connected by more than one hyperedge, which is fundamentally

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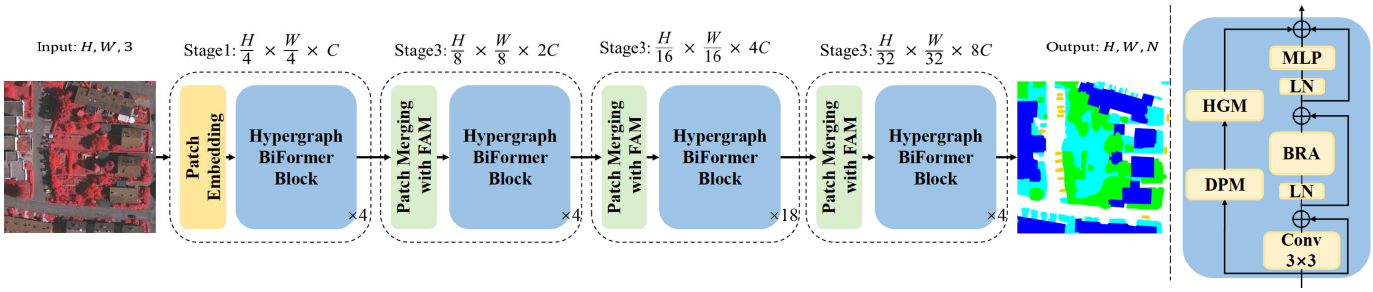


Fig. 2. (Left) Overall architecture of our HGBT. (Right) Details of an HGBT block.

different from a simple graph. This allows hypergraphs to associate nodes from various perspectives, greatly enhancing their capacity to represent the features. Meanwhile, it also empowers hypergraphs with the capability to model complex topological structures and extract high-order information.

1) Hypergraph Structural Definition: Suppose $G = (V, E)$ represents a simple graph, where V and E represent the vertices and edges, respectively. A simple graph can be generalized to a hypergraph, which can be defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of vertices and $\mathcal{E}$ is the set of hyperedges.

A hypergraph can also be simply represented as an incidence matrix $\mathbf{H}$. As presented in Fig. 3, the elements of the $\mathbf{H}$ matrix are set to 1 if the current vertex is connected to a hyperedge and set to 0 otherwise. The incidence matrix $\mathbf{H}$ can be defined with entries as follows:

$$h(v, e) = \begin{cases} 1, & \text{if } v \in e \\ 0, & \text{if } v \notin e. \end{cases} \tag{4}$$

2) Hypergraph Convolution: Whether a hyperedge connects two vertices can be determined by different ways, such as distance-based methods and attribute-based methods. In HGConv, we use the distance-based method to construct hyperedge sets. The distance-based method determines whether to establish a hyperedge between two vertices by calculating the distance of the correlation between each pair of point features. Specifically, distance d between any two vertices can be defined as follows:

$$d(v_i, v_j) = \|\mathbf{X}(v_i) - \mathbf{X}(v_j)\|_2 \tag{5}$$

where v_i and v_j are a pair of vertices whose feature distance is to be calculated, $\mathbf{X}(v_i)$ and $\mathbf{X}(v_j)$ are the features of v_i and v_j, and $\|\cdot\|_2$ is the Euclidean distance.

Therefore, we can get a distance matrix $\mathbf{Dis}(\mathbf{X}) \in \mathbb{R}^{N\times N}$, each row in $\mathbf{Dis}(\mathbf{X})$ represents the distance from one of the vertices to itself and the remaining $N-1$ vertices. After that, we apply the kNN algorithm to filter out the top-k vertices $\{v_1, v_2, \ldots, v_k\}$ with the largest distance to the current vertex v_i. Therefore, the hyperedge set can be expressed as follows:

$$\delta(v) = \left\{(v_i, v_j) | i \in 1, 2, \ldots, N, j \in 1, 2, \ldots, k\right\} \tag{6}$$

where N is the number of vertices on $\mathcal{G}$.

After constructing the incidence matrix $\mathbf{H}$ with p hyperedges based on $\delta(v)$, we can perform HGConv. The degree of any hyperedge and vertex can be calculated by

$$\begin{aligned} d(e) &= \Sigma_{v\in\mathcal{V}}\mathbf{H}(v, e) \\ d(v) &= \Sigma_{e\in\mathcal{E}}\mathbf{H}(v, e). \end{aligned} \tag{7}$$

The degree matrix of all vertices is represented as $\mathbf{D}_v$, and the degree matrix of all hyperedges is represented as $\mathbf{D}_e$.

As presented in the right of Fig. 3, for the feature of ith layer, $\mathbf{X}^i$ can be learned on the hypergraph structure by learning on vertices and on hyperedges and the next layer feature $\mathbf{X}^{i+1}$ can be calculated by

$$\mathbf{X}^{i+1} = \sigma\left(\mathbf{D}_v^{-1}\mathbf{H}\mathbf{W}\mathbf{D}_e^{-1}\mathbf{H}^T\mathbf{X}^i\mathbf{\Theta}^i\right) \tag{8}$$

where $\mathbf{W}$ represents the weight of hyperedges (which is set to a default value in this article) and $\mathbf{\Theta}$ is the learnable parameters.

D. Dual-Pooling Module

A challenging issue in Transformer design is that global self-attention is very expensive to compute, whereas local self-attention often limits the field of interactions of each token. BiFormer computes the attention map within a limited set of up to k-nearest neighboring regions, thereby significantly reducing the computational burden, even when compared with methods based on halo and shift operations or cross-attention mechanisms. However, in BiFormer, the expansion of the receptive field is relatively slow, requiring the stacking of a large number of neighboring regions selected by the k value to achieve global self-attention. This is because BiFormer tends to filter out regions that are most strongly correlated with themselves, resulting in the spatial structural information that has been ignored still not being selected, even with a small increase in the k value. However, this spatial information often plays a crucial role in removing classification errors in RS imagery caused by factors such as occlusion. A sufficiently large receptive field is essential for achieving high performance, particularly in HRRSIs semantic segmentation. Hence, it is crucial to efficiently obtain a large receptive field.

Hence, we propose the DPM, illustrated in Fig. 4, to capture pixel-level information by introducing and conducting soft pooling [24] and adaptive pooling [23] operations in the horizontal and vertical stripes in parallel, which handles the local spatial information better than existing schemes that rely solely on convolutions. DPM naturally supports arbitrary input resolutions and is thus especially effective and friendly for RS images.

In dual pooling, an input token feature $\mathbf{X} \in \mathbb{R}^{(HW)\times C}$ is initially restored to a 2-D spatial structure to adapt to subsequent pooling operations by passing through a convolutional layer with a kernel size of 3 and a dilation rate of 2. Following this initial step, we apply global soft pooling and adaptive pooling operations to extract pixel-level information in two orthogonal

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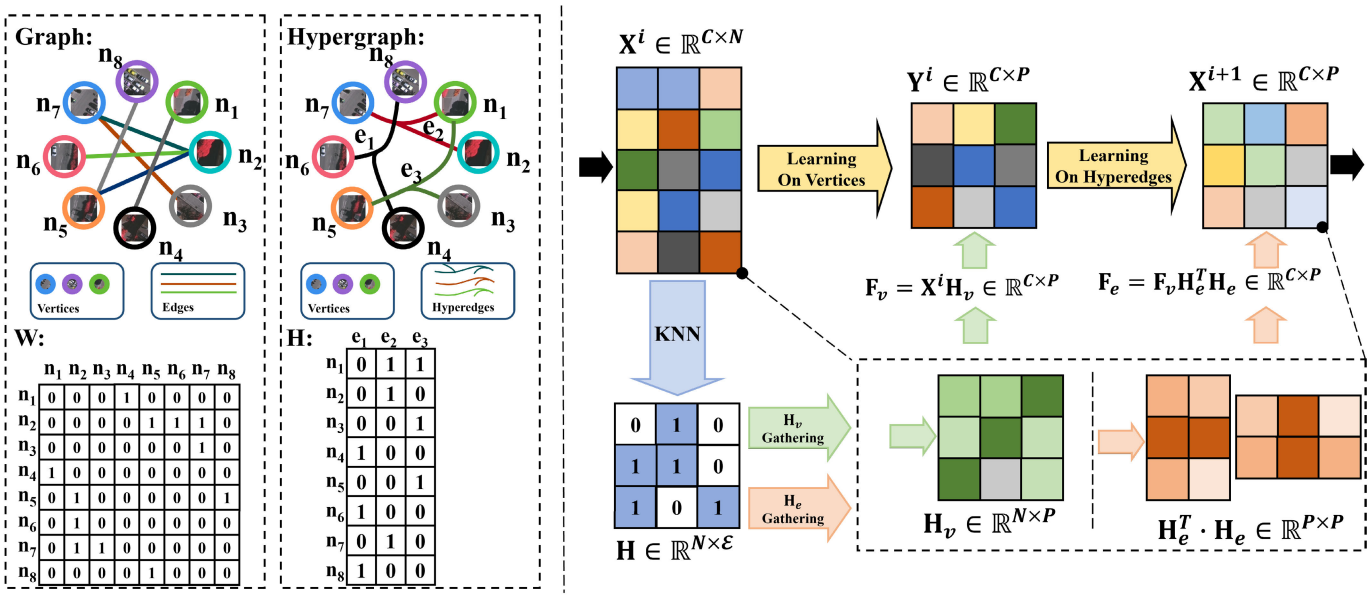


Fig. 3. (Left) Comparison between graph and hypergraph. (Right) Details of HGM.

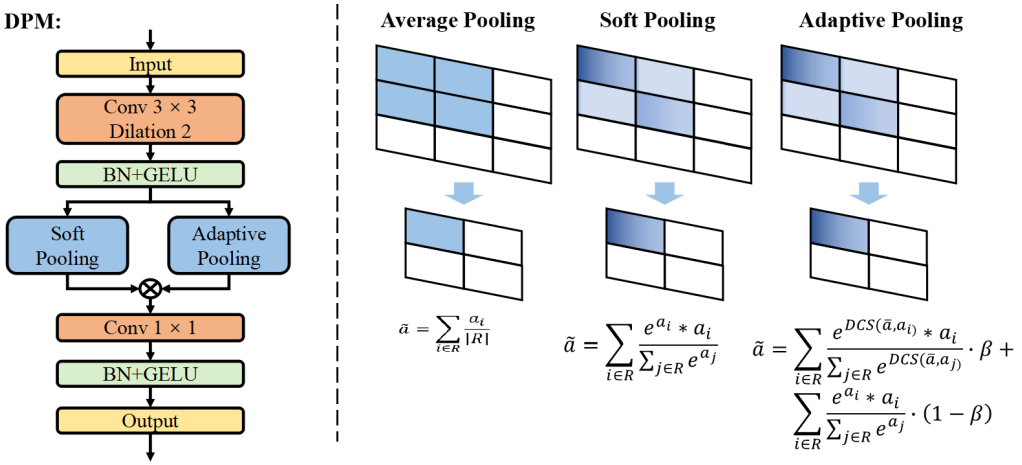


Fig. 4. (Left) Details of DPM. (Right) Comparison between different pooling methods.

directions within the spatial domain, namely, vertical and horizontal directions. Specifically, supposing $\mathbf{a}_i$ as a region to be pooled, the calculation formula of two exponential pooling operations can be defined as follows:

$$\widetilde{\mathbf{s}_h} = \sum_{i\in H} \frac{e^{\mathbf{a}_i} * \mathbf{a}_i}{\sum_{j\in W} e^{\mathbf{a}_j}}$$

$$\widetilde{a_v} = \sum_{i\in W} \frac{e^{\mathrm{DSC}(\bar{\mathbf{a}},\mathbf{a}_i)} * \mathbf{a}_i}{\sum_{j\in H} e^{\mathrm{DSC}(\bar{\mathbf{a}},\mathbf{a}_j)}} \cdot \beta + \widetilde{\mathbf{s}_v} \cdot (1-\beta) \tag{9}$$

$$\mathrm{DSC}(\bar{\mathbf{a}}, \mathbf{a}_i) = \frac{2 \cdot \left\|\bar{\mathbf{a}}, \mathbf{a}_i\right\|}{\left\|\bar{\mathbf{a}}\right\|^2 + \left\|\mathbf{a}_i\right\|^2} \tag{10}$$

where $\widetilde{\mathbf{s}_h} \in \mathbb{R}^{1\times W\times C}$ and $\widetilde{\mathbf{a}_v} \in \mathbb{R}^{H\times 1\times C}$ represents the resulting pooled region of soft pooling and adaptive pooling, respectively. $\bar{\mathbf{a}}$ represents the mean result and $\beta \in \{0, \ldots, 1\}$ is a weight mask.

By multiplying the pooling tensors from the vertical and horizontal directions, we obtain the position-related attention map $\mathbf{A}$, where $\mathbf{A} \in \mathbb{R}^{H\times W\times C}$. A 1×1 convolutional layer, followed by batch normalization and a GELU activation, is then applied to $\mathbf{A}$ to adjust its dimensions to align with the output. Supposing $\mathbf{X}_{\text{in}}$ as the restored feature, the output feature $\mathbf{X}_{\text{out}} \in \mathbb{R}^{(HW)\times C}$ can be summarized as follows:

$$\mathbf{X}_{\text{out}} = \varphi(\mathrm{SoftPool}(\mathbf{X}_{\text{in}}) \otimes \mathrm{AdaPool}(\mathbf{X}_{\text{in}})) \tag{11}$$

where $\varphi(\cdot)$ represents the last convolutional layer with batch normalization and GELU activation layer.

E. Feature Aggregation Module

The patch merging is used to down sample and reshape features. Swin Transformer achieves patch merging through the superposition of feature maps with displacements, followed by the linear layer processing. In contrast, BiFormer employs convolution with batch normalization for this purpose (see Fig. 5). Nonetheless, these approaches frequently lead to significant information loss, including fine-grained details and structural nuances, posing challenges to the semantic segmentation of RS images containing densely distributed and small-scale objects.

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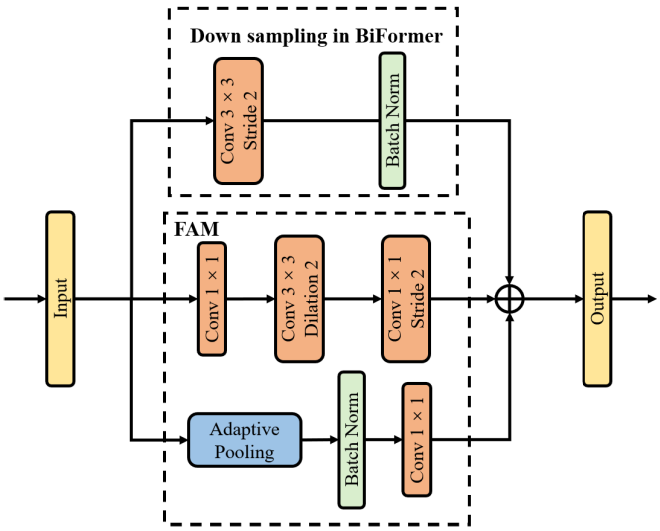


Fig. 5. Illustration of the patch merging with FAM.

Inspired by the inclusion of a local context enhancement term in the BiFormer block, we propose the FAM and integrate it with a novel patch merging strategy within the downsampling part in BiFormer to enhance the capability of the model to extract features from local content and small objects.

In summary, in the HGBT stage, patches to be merged are fed into down sampling and FAM in parallel. The down sampling is consistent with BiFormer and FAM is composed of a convolutional structure for enlarging the receptive field and an adaptive pooling structure for producing downsampled volumes that closely resemble the input volume. Sequential convolutional layers of 1×1 convolution, 3×3 convolution with a dilation rate of 2, and a 1×1 convolution with a stride of 2 are employed to capture a larger receptive field. The adaptive pooling operation can be calculated by (9).

Given an input feature $\mathbf{S}^i \in \mathbb{R}^{H\times W\times C}$ in ith stage, the result of downsampling and FAM can be defined as follows:

$$\begin{aligned}\mathbf{O}_D &= \mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathbf{S}^i))\\ \mathbf{O}_F &= \varphi(\mathrm{Conv}_{3\times3}(\varphi(\mathbf{S}^i))) \oplus \varphi(\mathrm{AdaPool}(\mathbf{S}^i))\end{aligned} \tag{12}$$

where $\mathrm{BN}(\cdot)$ represents the batch normalization layer.

Finally, the patch merging operation with the proposed FAM can be defined as follows:

$$\mathbf{S}^{i+1} = \mathbf{O}_D \oplus \mathbf{O}_F \tag{13}$$

where $\mathbf{S}^{i+1} \in \mathbb{R}^{(H/2)\times(W/2)\times 2C}$ is the output of merged patches.

IV. Experiments

A. Datasets

We evaluate our network on the ISPRS Vaihingen and Potsdam datasets, both of which are widely recognized benchmarks in the field of RS. The ISPRS Vaihingen dataset contains 33 images with six semantic classes labeled including cars, trees, buildings, impervious surfaces, low-vegetation, and backgrounds. The Potsdam dataset contains 38 images along with ground-truth annotations for six classes, including impervious surface, tree, building, low vegetation, car, and background. In order to make a fair comparison, we keep consistent with the previous work [60]. All images and their corresponding label masks are cropped into patches with 1024×1024 pixels. For the Vaihingen dataset, 16 images are used for training and the remaining 17 images are used for testing. For the Potsdam dataset, 24 images are used for training and 14 for testing. ISPRS benchmark provides two kinds of annotations for each image: labels and eroded labels. Eroded labels refer to labels that have undergone morphological erosion to refine the boundaries, reducing noise and uncertainties around the edges of labeled areas. This is particularly useful for improving the accuracy of image segmentation tasks.

B. Evaluation Metrics

We evaluate the performance of all the methods using three metrics: overall accuracy (OA), mIoU, and F_1. The overall performance of semantic segmentation methods is evaluated by OA, which does not consider the differences among individual classes. It simply calculates the accuracy of classifying all samples. mIoU and F_1 score are commonly used evaluation metrics for semantic segmentation methods and provide valuable insights into their performance, allowing for a comprehensive analysis of their effectiveness in handling different classes and achieving accurate segmentation results. These metrics can be defined as follows:

$$\begin{aligned}\mathrm{OA} &= \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}}\\ \mathrm{mIoU} &= \frac{1}{C}\sum_{i=1}^{C}\frac{\mathrm{TP}_i}{\mathrm{TP}_i+\mathrm{FP}_i+\mathrm{FN}_i}\\ F1 &= \frac{2\cdot P\cdot R}{P+R}\end{aligned} \tag{14}$$

where TP denotes the true positive, TN denotes the true negative, FP denotes the false positive, FN denotes the false negative, C represents the number of classes, P represents the precision, and R represents the recall.

C. Implementation Details

We implement our proposed network and train it on 2 NVIDIA Tesla V100 GPUs. The AdamW optimizer and Poly scheduler are adopted. We set the batch size to 8, the learning rate to 6e−5, and set the weight decay to 0.01. To alleviate the imbalance of the dataset categories, we use the joint loss function. All the experiments are trained for 150 epochs. To ensure fair comparisons, we maintain the same settings for all the methods we have reproduced.

D. Comparison With State-of-the-Art Methods

We compare the proposed HGBT with some competitive representative methods, including FCN [29], BoTNet [21], Deeplab V3+ [73], dual-attention network (DANet) [48], ST-UNet [61], EANet [41], ResT [74], CTMFNet [63], CMTFNet [54], and LSKNet [75]. FCN, BoTNet, Deeplab v3+, EANet, and LSKNet are all CNN-based methods, utilizing ResNet50 as their backbone. DANet is an attention-based method. ST-UNet, ResT, CTMFNet, and CMTFNet are

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TABLE I

RESULTS ON ISPRS VAIHINGEN DATASET FOR SEMANTIC SEGMENTATION FROM FIVE INDEPENDENT RUNS. THE EVALUATION METRICS INCLUDE CLASSWISE IOU FOR FIVE CATEGORIES, OA, MEAN IOU (MIOU), AVERAGE $F1$ SCORE (AVG.F1), AND p-VALUE IN PAIRED T-TESTS OF AVG.F1. (BOLD DENOTES THE BEST RESULTS AND "†" DENOTES STATISTICAL SIGNIFICANCE WITH p-VALUE <0.05 IN PAIRED T-TESTS)

Method	Class IoU (%)					Evaluation index (%)			
	Imp.Surf.	Tree	Building	Low Veg.	Car	OA	mIoU	Avg.F1	p value
FCN [29]	83.93	78.47	90.07	69.85	64.28	89.35±1.98	77.32±2.15	86.85±2.21	4.15e-3
BoTNet [21]	83.23	79.89	88.96	68.52	67.73	88.85±1.85	77.67±1.92	87.15±2.03	3.85e-3
DeepLab v3+ [73]	84.70	79.93	88.78	68.43	68.86	89.15±1.79	78.14±1.85	87.45±1.95	3.45e-3
EANet [41]	84.83	80.08	89.60	70.10	71.80	89.45±1.62	79.28±1.78	88.15±1.82	2.65e-3
DANet [48]	84.52	80.45	89.45	70.33	69.18	89.45±1.65	78.79±1.81	87.85±1.88	2.85e-3
ST-UNet [61]	84.90	79.88	89.71	70.06	71.77	89.45±1.51	79.27±1.74	88.15±1.79	2.45e-3
ResT [74]	84.99	80.29	89.57	69.87	72.39	89.55±1.48	79.42±1.71	88.25±1.75	2.25e-3
LSKNet [75]	**87.62**	**82.95**	91.65	**75.43**	75.85	**93.40±1.22**	**84.70±1.45**	**91.50±1.48**	1.25e-3
CTMFNet [63]	-	-	-	-	-	91.60	84.34	91.37	-
CMTFNet [54]	-	-	-	-	-	88.71	77.95	87.42	-
HGBT	86.25	80.62	**92.16**	72.35	**78.59**	91.07±1.15†	81.99±1.28†	89.96±1.31†	-

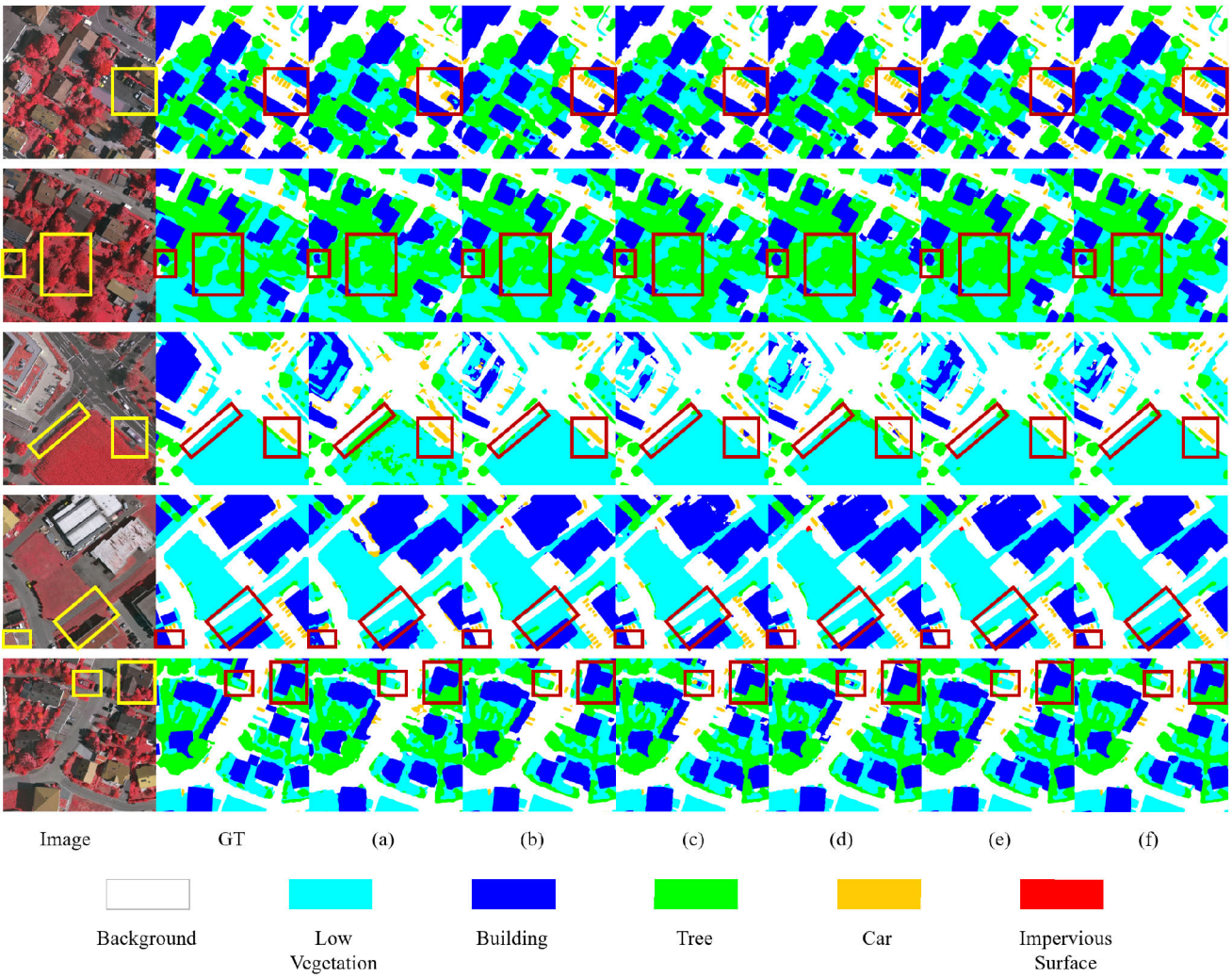


Fig. 6. Visualization of semantic segmentation results on ISPRS Vaihingen dataset for (a) FCN, (b) DeepLab v3+, (c) ST-UNet, (d) EANet, (e) LSKNet, and (f) Ours. The rectangles highlight regions where our method performs particularly better than others.

transformer-based methods. It is important to highlight that in the network architecture, both Transformer and CNN blocks are integrated into ST-Unet, CTMFNet, and CMTFNet, but they are U-Net shaped, whereas ours is hierarchical. For HGBT, due to considerations of computational efficiency, we have chosen BiFormer-S as the backbone, which possesses a lower memory requirement and computational overhead.

1) Results on Vaihingen Dataset: In Table I, the performance metrics for each semantic segmentation method are detailed. Evidently, the HGBT model demonstrates

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TABLE II

RESULTS ON ISPRS POTSDAM DATASET FOR SEMANTIC SEGMENTATION FROM FIVE INDEPENDENT RUNS. THE EVALUATION METRICS INCLUDE CLASSWISE IOU FOR FIVE CATEGORIES, OA, MEAN IOU (MIOU), AVERAGE $F1$ SCORE (AVG.F1), AND p-VALUE IN PAIRED T-TESTS OF AVG.F1. (BOLD DENOTES THE BEST RESULTS AND "†" DENOTES STATISTICAL SIGNIFICANCE WITH p-VALUE <0.05 IN PAIRED T-TESTS

Method	Class IoU (%)					Evaluation index (%)			
	Imp.Surf.	Tree	Building	Low Veg.	Car	OA	mIoU	Avg.F1	p value
FCN [29]	84.28	76.58	91.52	74.38	85.78	88.45±2.05	82.51±2.18	90.15±2.25	4.25e-3
BoTNet [21]	84.81	77.36	91.62	75.16	89.95	89.25±1.92	83.78±2.05	90.95±2.12	3.82e-3
DeepLab v3+ [73]	84.28	76.59	91.52	74.38	89.83	89.05±1.85	83.32±1.98	90.65±2.05	3.95e-3
EANet [41]	85.69	78.05	92.17	75.26	91.08	89.75±1.72	84.45±1.85	91.35±1.92	2.85e-3
DANet [48]	86.17	77.96	92.63	75.74	91.75	89.95±1.68	84.85±1.82	91.55±1.88	2.65e-3
ST-UNet [61]	85.32	77.57	91.98	75.23	90.65	89.55±1.65	84.15±1.78	91.15±1.85	3.15e-3
ResT [74]	86.57	78.69	92.76	75.92	91.46	92.25±1.52	85.08±1.65	91.75±1.72	2.45e-3
LSKNet [75]	**89.15**	**83.25**	93.85	77.65	92.10	92.0±1.25	**87.2±1.42**	**93.1±1.45**	1.15e-3
CTMFNet [63]	-	-	-	-	-	91.38	86.85	92.84	-
CMTFNet [54]	-	-	-	-	-	89.89	83.57	90.93	-
HGBT	88.00	80.50	**93.96**	**77.88**	**93.32**	**93.20±1.18**†	86.73±1.35†	92.76±1.38†	-

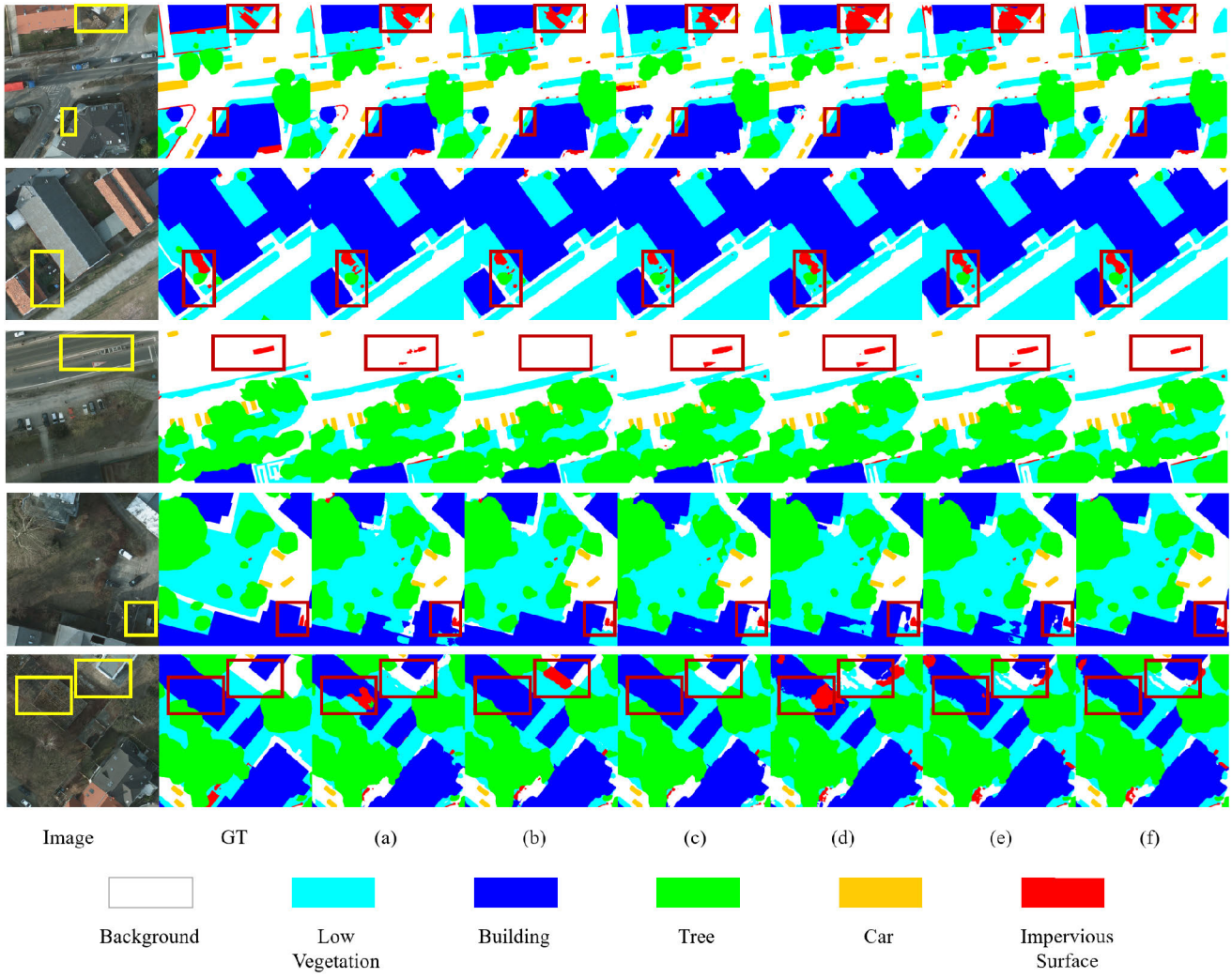


Fig. 7. Visualization of semantic segmentation results on ISPRS Potsdam datasets. The compared methods (a) FCN, (b) DeepLab v3+, (c) ST-UNet, (d) EANet, (e) LSKNet, and (f) Ours. The rectangles highlight regions where our method performs particularly better than others.

competitive performance against state-of-the-art methods, achieving superior results in both classwise IoU and overall metrics. Using dilated convolutions, Deeplab V3+, and a bottleneck structure, BoTNet enhances the receptive field for global context understanding, differing from DANet's dual-attention mechanism, which shows lower performance in global context modeling compared to our HGBT. Deeplab V3+ surpasses other CNN models through its atrous spatial pyramid pooling and advanced decoder design. ST-UNet, though slightly less effective than EANet, merges spatial

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details from encoder layers via skip connections, achieving better results in the "building" and "impervious surface" categories. Recent Transformer-based methods have shown promising results. LSKNet achieves strong performance through its large-scale kernel selective mechanism, particularly excelling in the "tree" and "building" categories. However, our HGBT demonstrates superior performance in challenging categories such as "car" (78.59% versus 75.85%), where fine-grained feature extraction is crucial. CTMFNet and CMTFNet, while introducing innovative architectural elements, show limitations in capturing the complex spatial relationships inherent in HRRSIs, as reflected in their overall performance metrics. Our proposed HGBT surpasses Deeplab V3+ with a notable improvement of 3.41% in mIoU and 2.11% in Avg.F1 and it achieves a 2.58% higher mIoU compared with ST-UNet. Despite the Swin transformer blocks' adeptness at global modeling, ST-UNet's sole reliance on this architecture falls short for HRRSIs. ResT employs efficient transformer blocks to slightly improve segmentation accuracy over ST-UNet, demonstrating the effectiveness of combining local information in transformer architectures. Notably, HGBT's hypergraph-based approach offers a more effective mechanism for modeling complex spatial relationships, leading to more robust performance across all categories. This is particularly evident in the challenging "car" and "low vegetation" classes, where HGBT achieves significant improvements over both traditional CNN-based methods and recent transformer-based approaches. The statistical significance of our results further validates the effectiveness of our proposed method.

Presented in Fig. 6 are the semantic segmentation performances from Table I, highlighting the superior performance of HGBT. LSKNet segmentation is observed to suffer from inadequate spatial detail representation, causing numerous semantic inconsistencies. In contrast, HGBT demonstrates superior accuracy in identifying similar features, significantly minimizing segmentation errors. It's significant to mention HGBT's proficiency in distinguishing "car" from "background" or "building," an aspect often challenging due to texture similarities that confuse other models. In addition, HGBT's capability to effectively process dense and small objects is evidenced in the first rows.

2) Results on Potsdam Dataset: To demonstrate the effectiveness of the proposed HGBT, we conducted further validation on the Potsdam dataset. Table II shows that our HGBT achieves an mIoU of 86.73% and an Avg.F1 score of 92.76%, outperforming most competing approaches. While LSKNet achieves competitive results with an mIoU of 87.2%, our HGBT demonstrates superior performance in challenging categories, particularly in "car" detection (93.32% versus 92.10%). CTMFNet and CMTFNet, despite their sophisticated architectures, achieve lower overall performance with mIoU scores of 86.85% and 83.57%, respectively. It is noteworthy that due to dataset variances, particularly the higher spatial resolution and more distinct object boundaries in Potsdam, the recognition accuracy for identical categories is consistently higher than in the Vaihingen dataset. This is evident across all methods, with improvements ranging from 3% to 5% in mIoU scores. Furthermore, for both datasets, the proposed HGBT significantly surpasses other methods in identifying the "car" category, showing remarkable consistency in handling small objects. This improvement is credited to the HGM module, which more effectively captures complex topological relationships with hypergraphs, thereby boosting detection precision for small-scale objects and those with similar appearances. The statistical significance of our results further validates the robustness of HGBT across different dataset characteristics and challenging scenarios.

We visualize the segmentation results in Fig. 7. In the first and second rows, we can note that "impervious surface" is often misclassified due to its adjacency to "low vegetation." Nevertheless, due to HGBT's improved capturing of local contextual information, it can still yield relatively accurate inferences in such scenarios. In the third and fifth rows, in contrast to other methods, HGBT successfully discriminates between "impervious surface" and "background," as well as "impervious surface" and "building," even in cases where they possess closely resembling color features. In the fourth row, our model demonstrates improved recognition of the "low vegetation" located above the "building."

Analyzing the segmentation results and p-values, it becomes evident that HGBT exhibits higher stability in its performance across the two datasets. This outcome highlights the effectiveness of our proposed HGBT methodology in achieving accurate and dependable semantic segmentation in HRRSIs.

3) Efficiency Analysis: For a comprehensive comparison, Table III presents detailed computational metrics including model parameters, FLOPs, inference speed (FPS), GPU memory consumption, and runtime measurements for both training and testing phases. As shown in Table III, the computational requirements vary significantly across different architectures. Traditional CNN-based methods like FCN demonstrate lightweight characteristics with faster inference speed and efficient training time. LSKNet requires substantial resources and relatively lower inference speed. In contrast, recent transformer-based approaches show higher computational demands. Due to the parallel design of HGBT, combining Biformer and Hypergraph, it achieves a balanced tradeoff with moderate parameter count and competitive inference speed. The testing time per image and training efficiency are comparable to other state-of-the-art methods. While these computational requirements may limit HGBT's deployment on resource-constrained devices, its superior segmentation performance and reasonable efficiency make for practical applications when accuracy is crucial.

E. Ablation Studies

To assess the effectiveness of the proposed network structure and its three key modules, we conduct ablation experiments using BiFormer-S as the baseline on the Potsdam dataset. In addition, we investigate the impact of the K value on the proposed HGM. In our HGBT model, the K value is set to ten during the ablation studies of the three modules, and the encoder uses the "small" configuration of BiFormer (i.e., the hidden size $C1 = 96$, the window size is 8, the number of layers for each stage is 2, 2, 6, and 2, and the number of heads per layer is 3, 6, 12, and 24).

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TABLE III

COMPARISON OF MODEL EFFICIENCY METRICS FOR SEMANTIC SEGMENTATION METHODS. THE EVALUATION METRICS INCLUDE MODEL PARAMETERS (M), COMPUTATIONAL COMPLEXITY (FLOPS), INFERENCE SPEED (FPS), GPU MEMORY CONSUMPTION (MEMORY), TRAINING TIME PER EPOCH, AND TESTING TIME PER IMAGE. ALL EXPERIMENTS ARE CONDUCTED ON A SINGLE NVIDIA TESLA V100 GPU WITH 32-GB MEMORY, USING INPUT IMAGES OF SIZE 1024 × 1024

Method	Params (M)	FLOPs (G)	FPS (f/s)	Memory (GB)	Train (min)	Test (s)
FCN [29]	14.72	89.5	35.2	4.8	12.5	0.15
BoTNet [21]	25.13	125.3	20.6	6.2	15.8	0.22
DeepLab v3+ [73]	34.48	156.7	25.4	7.5	16.5	0.25
EANet [41]	34.23	162.4	19.8	7.8	17.2	0.28
DANet [48]	66.56	198.5	16.5	9.2	19.5	0.32
ST-UNet [61]	160.97	245.8	11.6	12.5	24.8	0.45
ResT [74]	32.24	158.9	15.2	7.6	16.8	0.26
CTMFNet [63]	-	-	-	-	-	-
CMTFNet [54]	72.35	205.8	-	-	-	-
LSKNet [75]	40.45	228.6	19.2	11.5	23.6	0.42
HGBT	60.87	185.6	22.8	13.6	18.6	0.30

TABLE IV

RESULTS OF A DIFFERENT k VALUE IN KNN ALGORITHM

Hypergraph BiFormer	OA	mIoU	F1 Score
HGBT (k=3)	87.9	69.3	79.8
HGBT (k=10)	91.4	83.5	90.9
HGBT (k=20)	91.4	83.4	90.8
HGBT (k=30)	**91.4**	**83.7**	**91.0**

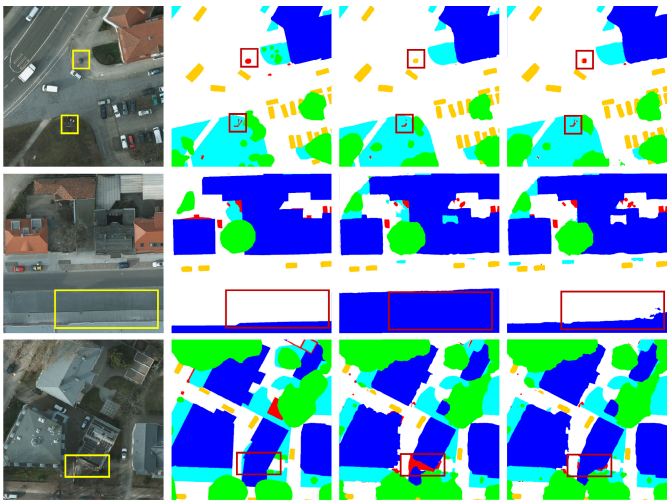


Fig. 8. Ablation study results before and after the inclusion of our proposed HGM. (a) Segmentation results without HGM. (b) Segmentation results with HGM.

1) Effect of k Value: Since the hypergraph is structured by kNN algorithm, different k values have a crucial effect on hypergraph learning. In Table IV, we explore the mIoU, OA and $F1$ score when $k = 3$, 10, 20, and 30. We find that $k = 30$ gives the best overall results, but it results in a substantial increase in computational complexity. Therefore, in our experiments, we choose $k = 3$, which offers slightly lower accuracy but requires less computational resources.

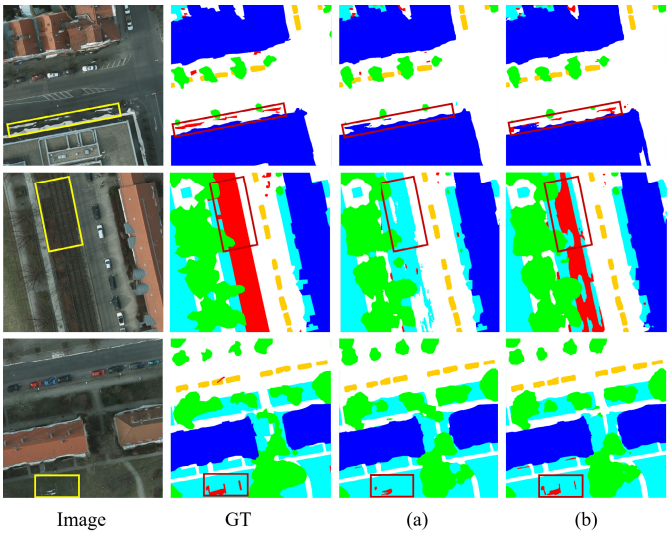


Fig. 9. Ablation study results before and after the inclusion of our proposed DPM. (a) Segmentation results without DPM. (b) Segmentation results with DPM.

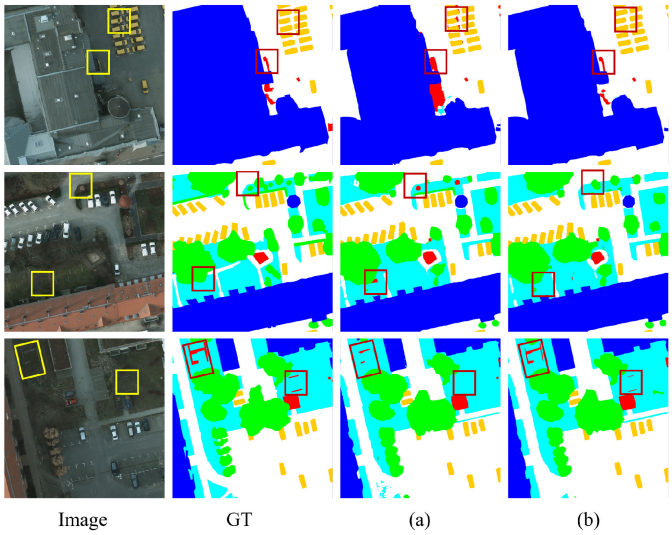


Fig. 10. Ablation study results before and after the inclusion of our proposed FAM. (a) Segmentation results without FAM. (b) Segmentation results with FAM.

2) Effect of HGM: Table V demonstrates a 1.85% improvement in mIoU and a 1.07% increase in Avg.F1 when incorporating the HGM into BiFormer. Notably, the segmentation accuracy for the "car" category shows the largest gain, with a 3.38% increase in IoU, followed by "low vegetation," which improved by 1.89%. A more intuitive comparison of the visual segmentation results is provided in Fig. 8. In the first row, the distinctive features of small-scale objects can be effectively segregated from neighboring objects prone to confusion after the integration of HGM. In the second and third rows, models without HGM incorrectly classify "building" as "impervious surface" and misidentify "background" as "building" due to their similar materials. However, our HGBT with the inclusion of HGM demonstrates a comparatively accurate assessment.

3) Effect of DPM: As shown in Table V, after using the DPM independently, the model achieves 1.54% and

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TABLE V

ABLATION STUDY RESULTS OF HGBT. THE EVALUATION METRICS INCLUDE CLASSWISE IOU FOR FIVE CATEGORIES AND THREE OVERALL METRICS (OA, MIOU, AND AVG.F1). RESULTS FOR OVERALL METRICS ARE PRESENTED AS MEAN ± STD FROM FIVE INDEPENDENT RUNS, WHERE "†" INDICATES STATISTICAL SIGNIFICANCE ($p < 0.05$) IN PAIRED T-TESTS

Model Name	Modules			IoU (%)					Evaluation index (%)		
	FAM	DPM	HGM	Imp.Surf.	Tree	Building	Low Veg.	Car	OA	mIoU	Avg.F1
Baseline				85.94	78.71	92.29	74.98	89.14	92.11±0.65	84.21±0.72	91.30±0.66
Baseline+FAM	✓			86.70	78.44	92.80	76.99	90.75	92.42±0.68	85.14±0.71	91.84±0.65
Baseline+DPM		✓		87.23	80.42	93.19	76.32	91.60	92.76±0.64	85.75±0.76	92.20±0.68
Baseline+HGM			✓	87.48	80.04	93.39	76.87	92.52	92.95±0.69	86.06±0.68	92.37±0.69
Baseline+FAM+DPM	✓	✓		87.50	80.11	94.01	77.17	92.55	93.11±0.72	86.27±0.67	92.49±0.72
Baseline+FAM+HGM	✓		✓	87.40	80.52	93.78	77.57	92.19	93.09±0.62	86.29±0.74	92.52±0.61
Baseline+DPM+HGM		✓	✓	87.95	**80.85**	**94.04**	77.69	92.00	93.13±0.65	86.50±0.70	92.64±0.67
HGBT	✓	✓	✓	**88.00**	80.50	93.96	**77.88**	**93.32**	**93.20±0.61**†	**86.73±0.68**†	**92.76±0.63**†

TABLE VI

RESULTS ON NOISE-AUGMENTED VAIHINGEN AND POTSDAM DATASETS. DIFFERENT LEVELS OF GAUSSIAN NOISE ($\sigma = 0.1$, 0.2, AND 0.3) ARE ADDED TO TEST THE MODEL ROBUSTNESS. THE EVALUATION METRICS INCLUDE MIOU (%) AND PERFORMANCE DECREASE COMPARED TO CLEAN IMAGES (DECREASE). BOLD DENOTES THE BEST RESULTS AND "†" DENOTES THE STATISTICAL SIGNIFICANCE WITH p-VALUE <0.05 IN PAIRED T-TESTS

Method	Vaihingen							Potsdam						
	σ=0.1		σ=0.2		σ=0.3		Clean	σ=0.1		σ=0.2		σ=0.3		Clean
	mIoU	Dec↓	mIoU	Dec↓	mIoU	Dec↓	mIoU	mIoU	Dec↓	mIoU	Dec↓	mIoU	Dec↓	mIoU
FCN [29]	72.85	-4.47	72.12	-5.20	67.56	-9.76	77.32	77.29	-5.22	76.31	-6.20	72.77	-9.74	82.51
DeepLab [73]	74.13	-4.01	71.25	-6.89	67.90	-10.24	78.14	79.10	-4.22	77.43	-5.89	75.03	-8.29	83.32
EANet [41]	77.15	-2.13	75.48	-3.80	72.95	-6.33	79.28	82.03	-2.42	78.79	-5.66	74.73	-9.72	84.45
DANet [48]	75.58	-3.21	72.89	-5.90	70.25	-8.54	78.79	81.78	-3.07	79.88	-4.97	77.64	-7.21	84.85
ST-UNet [61]	77.08	-2.19	75.35	-3.92	73.85	-5.42	79.27	81.06	-3.09	80.01	-4.14	77.98	-6.17	84.15
ResT [74]	77.25	-2.17	74.51	-4.91	74.02	-5.40	79.42	82.32	-2.76	81.14	-3.94	79.67	-5.41	85.08
LSKNet [75]	**82.45**	-2.25	78.85	-5.85	76.25	-8.45	84.70	84.76	-2.44	81.23	-5.97	78.72	-8.48	87.20
HGBT w/o HGM	78.88	-2.16	77.09	-3.95	75.22	-5.82	81.04	84.31	-1.96	82.07	-4.20	80.18	-6.09	86.27
HGBT	80.25†	**-1.74**	**79.69**†	**-2.30**	**77.85**†	**-4.14**	81.99	**85.12**†	**-1.61**	**84.56**†	**-2.17**	**82.85**†	**-3.88**	86.73

0.90% improvements on mIoU and Avg.F1, respectively. The effectiveness of DPM in our network is verified. Since the categories "car" and "low vegetation" are easily occluded by "tree" in RS images, extracting and recognizing their semantic features are difficult. The model after using the DPM increases by 2.46% in the category "car," 1.34% in the "low vegetation," and 1.71% in the "tree." As shown in the first row of Fig. 9.

As indicated in Table V, utilizing the DPM independently results in a 1.54% improvement in mIoU and a 0.90% increase in Avg.F1, demonstrating the effectiveness of DPM in our network. Categories like "car" and "low vegetation" are often occluded by "tree" in RS images, making it challenging to extract and recognize their semantic features. After applying DPM, the model shows a 2.46% increase in the "car" category, a 1.34% improvement for "low vegetation," and a 1.71% gain for "tree." The first row of Fig. 9 provides a visual comparison. The DPM effectively prevents segmentation errors of "impervious surface" and "background" caused by lighting changes and overlapping positions. In the second and third rows, the resemblance in appearance between "low vegetation" and "impervious surface" becomes apparent. Due to their frequent co-occurrence, instances of "impervious surface" are prone to being erroneously classified as "low vegetation." However, through the aggregation of distinctive features from both global context and local attributes, the HGBT model sustains the ability to make reasonably accurate inferences, even within such challenging scenarios. The visualization outcomes presented in Fig. 9 demonstrate that the incorporation of DPM effectively mitigates the adverse impact of mutual object occlusion. The result in Fig. 9 underscores the positive influence of introducing DPM in alleviating the detrimental effects arising from the occlusion of objects.

4) Effect of FAM: As shown in Table V, incorporating FAM into the BiFormer framework leads to a 0.93% increase in mIoU and a 0.54% improvement in Avg.F1. Based on the numerical results in Table V, FAM has the most significant impact on the "low vegetation" category, with a 2.01% IoU increase. The visual comparison of results is presented in Fig. 10. In the first and second rows, following the incorporation of FAM, small-scale objects situated in corner regions, such as "Tree" nestled in corners, "background" and the periphery of "car," can be adeptly segmented into their accurate categories. In the third row of results, our model demonstrates notable improvements in recognizing the elongated "impervious surface" region, which is situated between the areas of "low vegetation."

In addition, the combined effect of the modules within the BiFormer framework is analyzed, as shown in Table V. When both HGM and DPM are introduced together, mIoU improves by 2.29% and Avg.F1 by 1.34%. Including both HGM and FAM results in a 2.08% mIoU increase and a 1.22% rise in Avg.F1. With the combination of DPM and FAM, mIoU increases by 2.06%, while Avg.F1 gains 1.19%. Our HGBT, which incorporates the three key modules (HGM, FAM, and DPM), achieves a 2.52% improvement in mIoU and a 1.46% increase in Avg.F1 compared to BiFormer.

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The ablation study results have been rigorously validated through statistical testing. Through paired t-tests across five independent runs, we observe that each proposed component contributes significantly to the model's performance. The statistical analysis confirms that both individual modules and their combinations demonstrate significant improvements over the baseline architecture. These results provide strong statistical evidence for the effectiveness of our modular design, with all performance gains being statistically significant.

5) Robustness Analysis Under Image Quality Degradation: To thoroughly investigate HGBT's potential in real-world scenarios with varying image quality conditions, we conduct comprehensive experiments on degraded datasets. Specifically, we construct noise-augmented versions of both Vaihingen and Potsdam datasets by adding different levels of Gaussian noise ($\sigma = 0.1$, 0.2, and 0.3), simulating common image quality issues in practical RS applications.

As shown in Table VI, on the Vaihingen dataset, HGBT demonstrates remarkable robustness as noise levels increase. While LSKNet shows superior performance under mild noise conditions, its performance drops significantly at higher noise levels. In contrast, HGBT maintains better stability and even outperforms LSKNet under moderate and severe noise conditions ($\sigma = 0.2$ and 0.3). This superior noise resistance can be attributed to HGM's powerful capability in modeling high-order relationships. The substantial accuracy degradation of LSKNet might be due to its larger kernel size, which makes it more susceptible to noise interference and compromises its recognition ability in real-world scenarios. For the Potsdam dataset, HGBT consistently outperforms LSKNet across all noise levels, with significantly smaller performance decreases. This more pronounced difference compared to Vaihingen might be attributed to Potsdam's higher initial image quality, making models more sensitive to noise perturbations. The enhanced robustness of HGBT can be primarily attributed to its architectural design: the HGM maintains stable structural relationships even under noise perturbations, while the BiFormer component provides complementary global context that helps preserve essential semantic information.

V. Conclusion

In this work, we propose a novel Transformer-based network called HGBT, designed to capture both global and local contextual information simultaneously in HRRSIs. We construct an HGBT with a hierarchical structure by incorporating the proposed HGM, DPM, and FAM. Specifically, for the first time, we introduced the hypergraph structure into Transformer architecture. The proposed HGM can better model the local topology and establish high-order level information exchange, alleviating the limitation of discarding a significant portion of weakly correlated patches in BiFormer. In addition, DPM and FAM are proposed to further enhance the model's ability to recognize local information and small-scale features through the adaptive pooling and soft pooling. The DPM employs selective aggregation of significant local information in two spatial dimensions without resulting in high computation. The FAM obtains supplementary information amenable to dense prediction tasks that may be disregarded during the downsampling stage. We conduct an extensive ablation study and evaluate the performance of HGBT in ISPRS Vaihingen and Potsdam datasets to demonstrate the superiority of the proposed method in comparison with the state-of-the-art methods.

A limitation of HGBT is its inability to effectively segment mutually occluded objects. This shortcoming is primarily evident in the segmentation results, which often fail to accurately replicate the exact shape of ground objects, leading to jagged or irregular boundary lines. In the future, we plan to perform a hypergraph with more information exchange and explore methods of integrating point cloud data to address overlap limitations.

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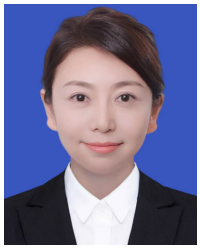
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